

Настройка сетевых сервисов DNS и DHCP

Лабораторная работа №8

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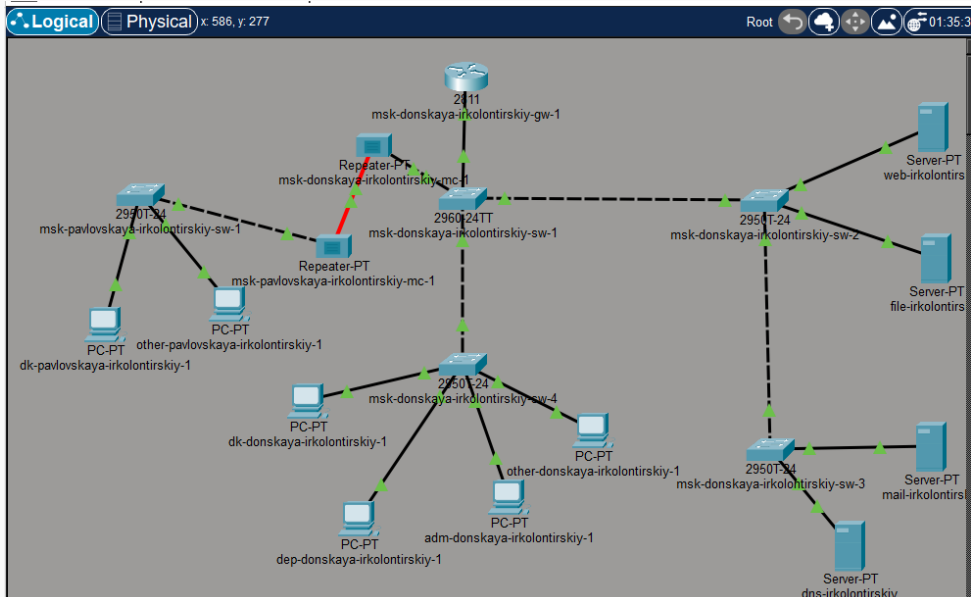
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Цель работы

Приобретение практических навыков настройки сетевых сервисов:

- DNS (Domain Name System)
- DHCP (Dynamic Host Configuration Protocol)

- Добавить и настроить DNS-сервер
- Настроить DHCP на маршрутизаторе
- Перевести клиентов на динамическую адресацию
- Проверить связность сети
- Изучить процесс DHCP в режиме симуляции



Настройка DNS-сервера

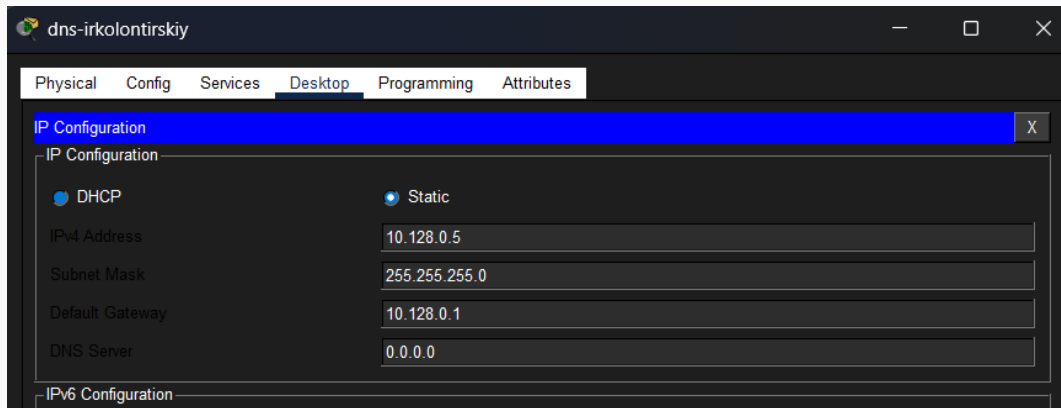
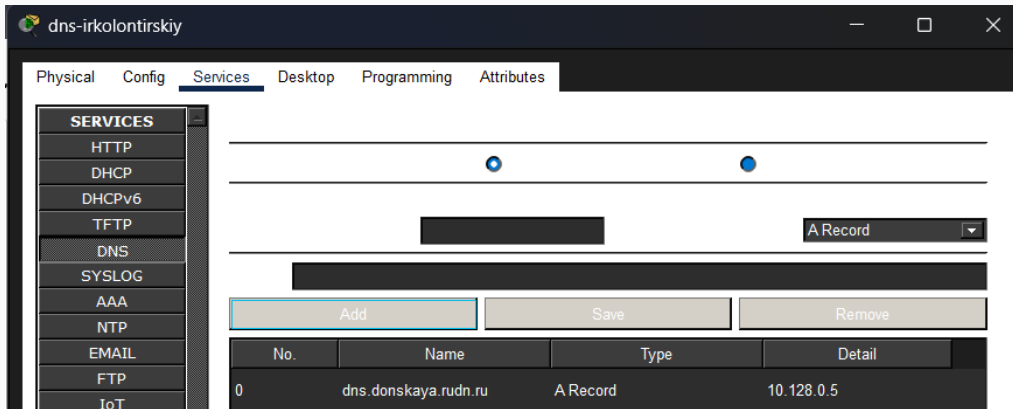


Рис. 2: Настройка IP DNS-сервера

- www.donskaya.rudn.ru → 10.128.0.2
- file.donskaya.rudn.ru → 10.128.0.3
- mail.donskaya.rudn.ru → 10.128.0.4
- dns.donskaya.rudn.ru → 10.128.0.5



The screenshot shows the dnsmasq web interface with the 'Services' tab selected. On the left, a sidebar lists various services, with 'DNS' highlighted. The main area contains a form for adding a new DNS record. The 'Name' field is empty, and the 'Type' dropdown is set to 'A Record'. Below the form are 'Add', 'Save', and 'Remove' buttons. At the bottom, a table displays the current DNS records.

No.	Name	Type	Detail
0	dns.donskaya.rudn.ru	A Record	10.128.0.5

Настройка DHCP

Параметры DHCP

- шлюз: .1
- DNS: 10.128.0.5
- исключены адреса:
 - .1-.29
 - .200-.254

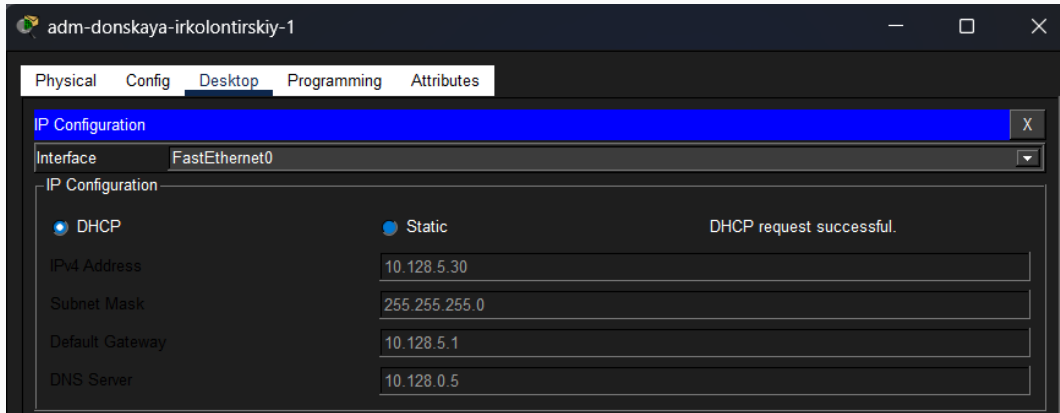
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Password:
msk-donskaya-irkolontirskiy-gw-1>en
Password:
msk-donskaya-irkolontirskiy-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-irkolontirskiy-gw-1(config)#
msk-donskaya-irkolontirskiy-gw-1(config)#ip name-server 10.128.0.5
msk-donskaya-irkolontirskiy-gw-1(config)#service dhcp
msk-donskaya-irkolontirskiy-gw-1(config)#ip dhcp pool dk
msk-donskaya-irkolontirskiy-gw-1(dhcp-config)#network 10.128.3.0 255.255.255.0
msk-donskaya-irkolontirskiy-gw-1(dhcp-config)#default-router 10.128.3.1
msk-donskaya-irkolontirskiy-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-irkolontirskiy-gw-1(dhcp-config)#exit
msk-donskaya-irkolontirskiy-gw-1(config)#ip dhcp excluded-address 10.128.3.1 10.128.3.29
msk-donskaya-irkolontirskiy-gw-1(config)#ip dhcp excluded-address 10.128.3.200 10.128.3.254
msk-donskaya-irkolontirskiy-gw-1(config)#ip dhcp pool departments
msk-donskaya-irkolontirskiy-gw-1(dhcp-config)#network 10.128.4.0 255.255.255.0
msk-donskaya-irkolontirskiy-gw-1(dhcp-config)#default-router 10.128.4.1
msk-donskaya-irkolontirskiy-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-irkolontirskiy-gw-1(dhcp-config)#exit
msk-donskaya-irkolontirskiy-gw-1(config)#ip dhcp excluded-address 10.128.4.1 10.128.4.29
msk-donskaya-irkolontirskiy-gw-1(config)#ip dhcp excluded-address 10.128.4.200 10.128.4.254
msk-donskaya-irkolontirskiy-gw-1(config)#ip dhcp pool adm
msk-donskaya-irkolontirskiy-gw-1(dhcp-config)#network 10.128.5.0 255.255.255.0
msk-donskaya-irkolontirskiy-gw-1(dhcp-config)#default-router 10.128.5.1
msk-donskaya-irkolontirskiy-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-irkolontirskiy-gw-1(dhcp-config)#exit
msk-donskaya-irkolontirskiy-gw-1(config)#ip dhcp excluded-address 10.128.5.1 10.128.5.29
msk-donskaya-irkolontirskiy-gw-1(config)#ip dhcp excluded-address 10.128.5.200 10.128.5.254
```

Проверка DHCP

Информация о пулах

- Пулы созданы корректно
- Адреса распределяются автоматически
- Есть активные выдачи



Выданные адреса

- 10.128.3.30, 10.128.3.31
- 10.128.4.30
- 10.128.5.30
- 10.128.6.30, 10.128.6.31

```
msk-donskaya-irkolontirskiy-gw-1#sh ip dhcp pool

Pool dk :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)          : 0 / 0
Total addresses                   : 254
Leased addresses                  : 2
Excluded addresses                : 8
Pending event                    : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
10.128.3.1         10.128.3.1      - 10.128.3.254      2 / 8 / 254

Pool departaments :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)          : 0 / 0
Total addresses                   : 254
Leased addresses                  : 1
Excluded addresses                : 8
Pending event                    : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
10.128.4.1         10.128.4.1      - 10.128.4.254      1 / 8 / 254

Pool adm :
Utilization mark (high/low)      : 100 / 0
Subnet size (first/next)          : 0 / 0
Total addresses                   : 254
Leased addresses                  : 1
Excluded addresses                : 8
Pending event                    : none
```

Клиенты DHCP

- Клиенты переведены на DHCP
- Автоматически получены:
 - IP-адрес
 - шлюз
 - DNS

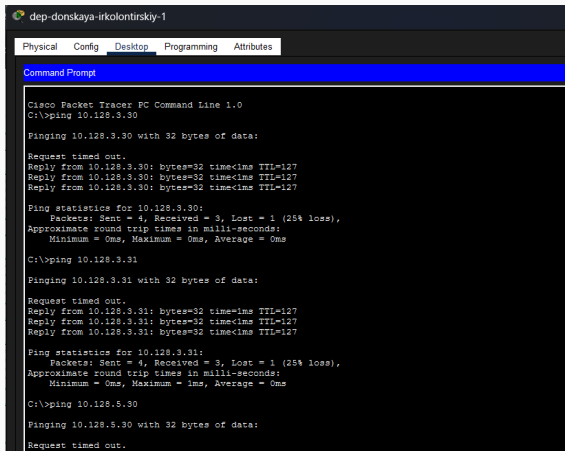
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msh-donskaya-irkolontirskiy-gw-1#  
msh-donskaya-irkolontirskiy-gw-1#sh ip dhcp bin
```

IP address	Client-ID/ Hardware address	Lease expiration	Type
10.128.3.30	00D0.9722.651D	--	Automatic
10.128.3.31	0009.7C6C.B04B	--	Automatic
10.128.4.30	0060.5C6C.E642	--	Automatic
10.128.5.30	000A.410A.37B2	--	Automatic
10.128.6.30	0001.649D.3CA8	--	Automatic
10.128.6.31	000C.CFE3.E457	--	Automatic

```
msh-donskaya-irkolontirskiy-gw-1#
```

Проверка связности

- Выполнен ping между подсетями
- Связность подтверждена
- Первый пакет теряется (ARP)



```
dep-donskaya-irkolontirskiy-1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.3.30

Pinging 10.128.3.30 with 32 bytes of data:

Request timed out.
Reply from 10.128.3.30: bytes=32 time<1ms TTL=127
Reply from 10.128.3.30: bytes=32 time<1ms TTL=127
Reply from 10.128.3.30: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.3.30:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.3.31

Pinging 10.128.3.31 with 32 bytes of data:

Request timed out.
Reply from 10.128.3.31: bytes=32 time=1ms TTL=127
Reply from 10.128.3.31: bytes=32 time<1ms TTL=127
Reply from 10.128.3.31: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.3.31:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.128.5.30

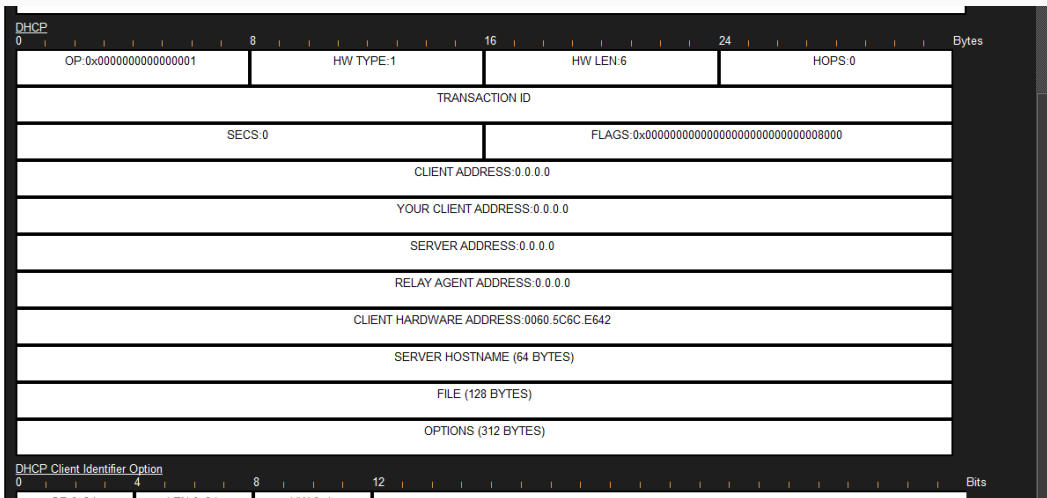
Pinging 10.128.5.30 with 32 bytes of data:

Request timed out.
```

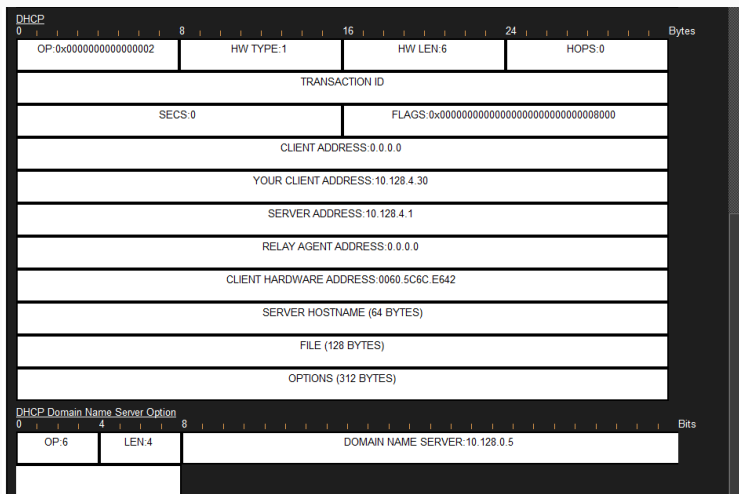
Работа DHCP (симуляция)

- Discover
- Offer
- Request
- ACK

- Клиент отправляет broadcast
- IP = 0.0.0.0

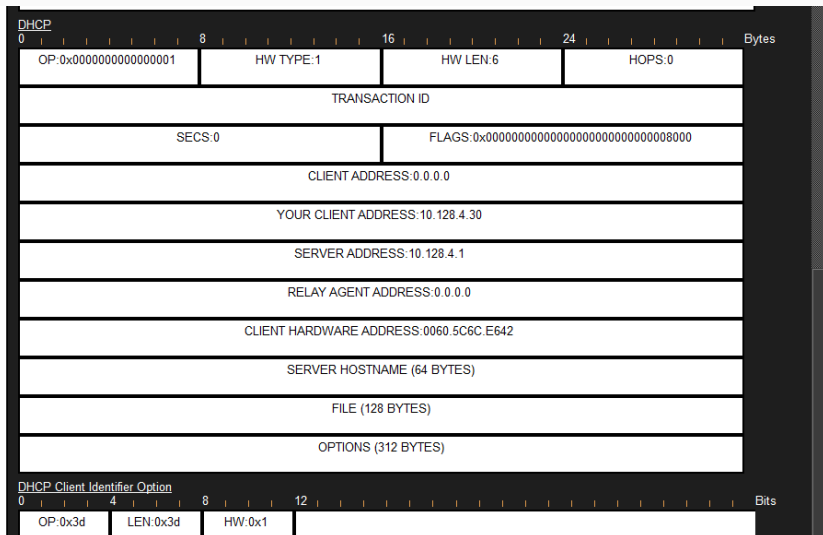


- Сервер предлагает IP
- Пример: 10.128.4.30

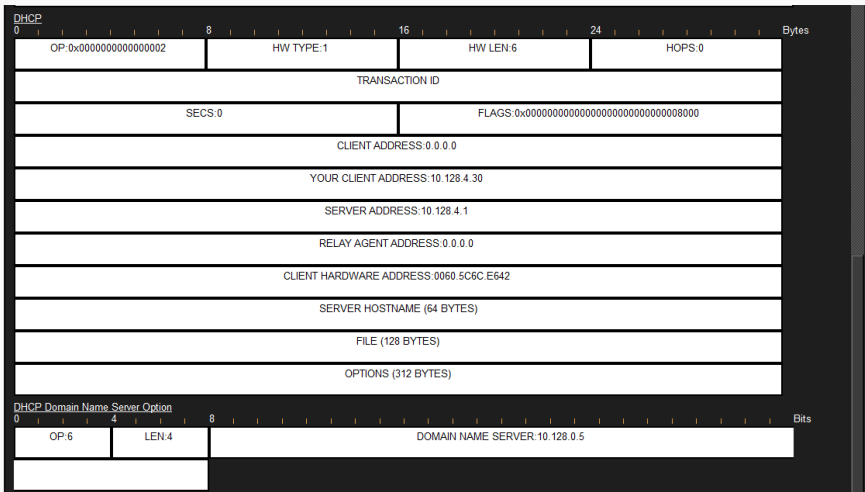


Request

- Клиент подтверждает выбор



- Сервер закрепляет адрес
- Передаются параметры сети



Вывод

- Настроен DNS-сервер
- Реализовано разрешение имён
- Настроен DHCP для нескольких подсетей
- Клиенты переведены на автонастройку
- Проверена работоспособность сети